

Proposal in Response to RFQ – Solar-Powered Irrigation Systems

Submitted to: Alliance for a Green Revolution in Africa (AGRA)

Submitted by: GreenTech Irrigation Solutions Ltd.

Date: October 5, 2025

Page 1 – Executive Summary

GreenTech Irrigation Solutions Ltd. (GTIS) is pleased to submit this quotation in response to AGRA's Request for Quotation (RFQ) dated September 2025 for the provision of solar-powered irrigation systems in Kenya, Ghana, and Tanzania.

Our company brings over **8 years of experience** in designing, installing, and maintaining renewable energy-powered agricultural systems across sub-Saharan Africa. We propose a solution that is:

- **Affordable:** Scaled packages starting from 0.5-acre to 5-acre irrigation systems.
- **Sustainable:** 100% solar-powered with minimal maintenance requirements.
- **Scalable:** Designed to grow with community needs.
- **Farmer-centric:** Includes comprehensive training and ongoing after-sales support.

We recognize AGRA's mission to strengthen smallholder farmers' resilience and productivity, and our proposal directly aligns with that vision.

Page 2 – Technical Proposal

System Design

Our proposed solar-powered irrigation solution includes:

- **Solar PV Array (1.5kW–5kW)** tailored to site conditions.
- **DC Submersible Pump** with 5,000–10,000 L/day capacity.
- **Drip Irrigation Network** optimized for smallholder crops (maize, vegetables, legumes).
- **Water Storage Tanks** (2,000–5,000 liters).
- **Smart Controller (optional upgrade)** to monitor water usage and pump efficiency.

Suitability and Innovation

- Modular design allows systems to serve 5–20 farmers per site.
- Option for remote monitoring via GSM-enabled smart controllers.
- Spare parts designed for easy replacement using locally available tools.

Adaptability to Rural Contexts

- Systems require no grid connection.
- Pump operation guaranteed at **70% performance even during cloudy conditions**.
- Training designed for users with low literacy through pictorial guides.

Gap: Site-specific solar irradiation data is pending and would be incorporated during the detailed feasibility stage.

Page 3 – Deliverables & Work Plan

Deliverables

1. **Supply and Installation** of solar-powered irrigation systems in 12 pilot sites (4 per country).
2. **Training Sessions** for at least 500 smallholder farmers (200 in Kenya, 150 in Ghana, 150 in Tanzania).
3. **Localized Training Materials** in Swahili, Twi, and English.
4. **Sustainability & After-Sales Support Plan**, including 24-month service guarantee.
5. **Quarterly Monitoring Reports** covering:
 - Pump performance
 - Water usage efficiency
 - Farmer adoption rates

Work Plan & Timeline

Activity	Month 1	Month 2	Month 3	Month 4	Month 5	Month 6
Procurement of equipment	X					
Site preparation	X	X				
Installation		X	X			
Farmer training			X	X		
Monitoring & Reporting				X	X	X

Gap: Final delivery timelines depend on import clearance procedures, which vary by country.

Page 4 – Company Experience & References

Company Profile

GreenTech Irrigation Solutions Ltd. is a registered company headquartered in Nairobi, Kenya, with satellite offices in Accra, Ghana, and Dar es Salaam, Tanzania.

Core Capabilities:

- Solar engineering and design
- Agricultural water management
- Farmer training and capacity building
- Local supply chain partnerships

Relevant Projects

- **Project A (2022–2023):** Installation of 15 solar irrigation systems in Kitui County, Kenya (served 350 farmers).
- **Project B (2021):** Partnership with Ministry of Food and Agriculture, Ghana, to supply 10 pilot irrigation units.
- **Project C (2020–2022):** Solar pump distribution initiative in Northern Tanzania.

References (partial – full contacts available on request)

1. [Name withheld], Ministry of Agriculture, Kenya – Solar Pump Initiative, 2023.
2. [Name withheld], NGO partner in Ghana – Community Irrigation Pilot, 2021.
3. [Name withheld], FAO project lead, Tanzania – Solar Water Program, 2020–2022.

Gap: Some client contact details pending confirmation for disclosure.

Page 5 – Financial Proposal & Sustainability

Financial Proposal (Indicative, in USD)

Item	Unit Cost (USD)	Quantity	Subtotal (USD)
Solar Irrigation System (average package)	4,500	12	54,000
Farmer Training (per participant)	30	500	15,000
Spare Parts Kit (per system)	300	12	3,600
Monitoring & Reporting (annual)	10,000	1	10,000
Total (Excl. Taxes & Freight)			82,600

Gap: Import duties, VAT, and logistics costs are excluded and will vary per country.

Sustainability & After-Sales Support

- Local technician training in each country for first-line repairs.
 - Service contracts available beyond warranty period at subsidized rates.
 - Spare parts to be stocked in Nairobi and Accra, with supply chains extending to rural areas.
 - Encouragement of farmer cooperatives to pool resources for system upkeep.
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Conclusion

GTIS is confident in its ability to deliver cost-effective, innovative, and sustainable irrigation systems that align with AGRA's objectives. While some details (site-specific data, exact import costs, and client disclosure permissions) remain pending, we are committed to addressing these gaps in the next stage of engagement.

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